Bertram's emphasis on kin selection as the prime mover of cooperation in lions has been challenged by C. Packer and A. Pusey. They claim that in many prides the two male lions are not related. Packer and Pusey suggest that reciprocal altruism is at least as likely as kin selection as an explanation for cooperation in lions. Probably both sides are right. Chapter 12 emphasizes that reciprocation ('Tit for Tat') can evolve only if a critical quorum of reciprocators can initially be mustered. This ensures that a would-be partner has a decent chance of being a reciprocator. Kinship is perhaps the most obvious way for this to happen. Relatives naturally tend to resemble one another, so even if the critical frequency is not met in the population at large it may be met within the family. Perhaps cooperation in lions got its start through the kin-effects suggested by Bertram, and this provided the necessary conditions for reciprocation to be favoured. The disagreement over lions can be settled only by facts, and facts, as ever, tell us only about the particular case, not the general theoretical argument.

p. 137 If C is my identical twin …

It is now widely understood that an identical twin is theoretically as valuable to you as you are to yourself—as long as the twin really is guaranteed identical. What is not so widely understood is that the same is true of a guaranteed monogamous mother. If you know for certain that your mother will continue to produce your father's children and only your father's children, your mother is as genetically valuable to you as an identical twin, or as yourself. Think of yourself as an offspring-producing machine. Then your monogamous mother is a (full) sibling-producing machine, and full siblings are as genetically valuable to you as your own offspring. Of course this neglects all kinds of practical considerations. For instance, your mother is older than you, though whether this makes her a better or worse bet for future reproduction than you yourself depends on particular circumstances—we can't give a general rule.

That argument assumes that your mother can be relied upon to continue producing your father's children, as opposed to some other male's children. The extent to which she can be relied upon depends upon the mating system of the species. If you are a member of a habitually promiscuous species, you obviously cannot count on your mother's offspring being your full siblings. Even under ideally monogamous conditions, there is one apparently inescapable consideration that tends to make your mother a worse bet than you are yourself. Your father may die. With the best will in the world, if your father is dead your mother can hardly be expected to go on producing his children, can she?

Well, as a matter of fact she can. The circumstances under which this can happen are obviously of great interest for the theory of kin selection. As mammals we are used to the idea that birth follows copulation after a fixed and rather short interval. A human male can father young posthumously, but not after he has been dead longer than nine months (except with the aid of deep-freezing in a sperm-bank). But there are several groups of insects in which a female stores sperm inside herself for the whole of her life, eking it out to fertilize eggs as the years go by, often long years after the death of her mate. If you are a member of a species that does this, you can potentially be really very sure that your mother will continue to be a good 'genetic bet'. A female ant mates only in a single mating flight, early in her life. The female then loses her wings and never mates again. Admittedly, in many ant species the female mates with several males on her mating flight. But if you happen to belong to one of those species whose females are always monogamous, you really can regard your mother as at least as good a genetic bet as you are yourself. The great point about being a young ant, as opposed to a young mammal, is that it doesn't matter whether your father is dead (indeed, he almost certainly *is* dead!). You can be pretty sure that your father's sperm are living on after him, and that your mother can continue to make full siblings for you.

It follows that, if we are interested in the evolutionary origins of sibling care and of phenomena like the insect soldiers, we should look with special attention towards those species in